

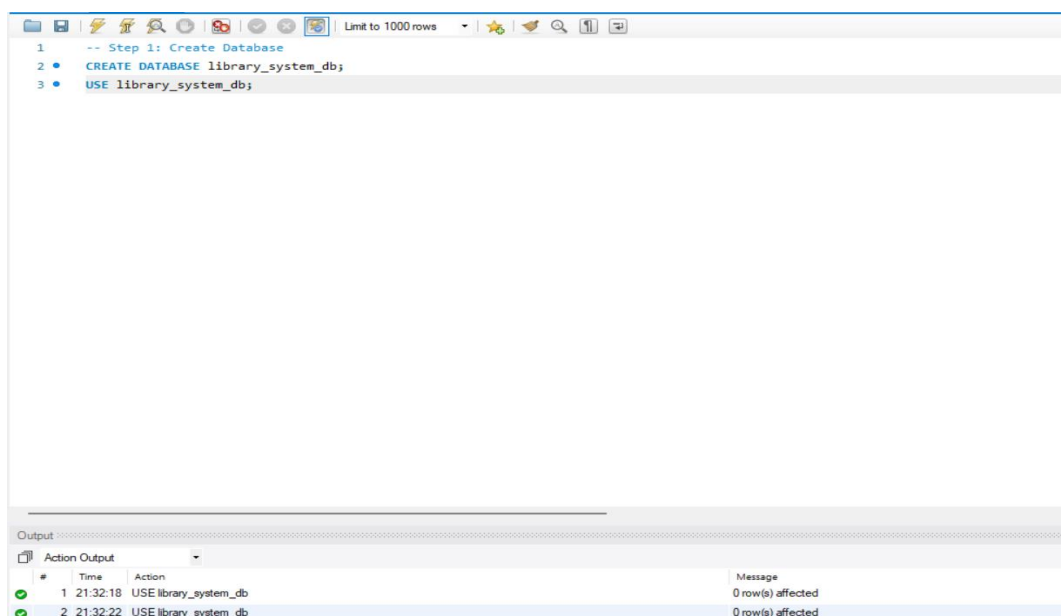
Week 1

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Question :

A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called **BookFlow**. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named **bookflow_db**. Then, design two tables: a **Books** table containing the columns **book_id**, **title**, **isbn**, and **published_year**, and a **Members** table containing the columns **member_id**, **full_name**, and **email**. Ensure data integrity by setting **book_id** and **member_id** as **PRIMARY KEYS**, applying a **NOT NULL** constraint to the **title** column so that every book must have a name, enforcing **UNIQUE** constraints on both **isbn** and **email** to prevent duplicate entries, and adding a **CHECK** constraint to ensure that the **published_year** is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the **Books** and **Members** tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as **CREATE DATABASE**, **CREATE TABLE**, and **INSERT** statements.

Step 1 — Create Database



```
-- Step 1: Create Database
1 CREATE DATABASE library_system_db;
2
3 USE library_system_db;
```

#	Time	Action	Message
1	21:32:18	USE library_system_db	0 row(s) affected
2	21:32:22	USE library_system_db	0 row(s) affected

Limit to 1000 rows

```
1 -- Step 1: Create Database
2 CREATE DATABASE library_system_db;
3 USE library_system_db;
4 -- Step 2: Create Books Table
5 CREATE TABLE Books(
6     book_id INT PRIMARY KEY,
7     title VARCHAR(100) NOT NULL,
8     isbn VARCHAR(20) UNIQUE,
9     published_year INT CHECK(published_year < 2027)
10 );
```

Output

#	Time	Action	Message
1	21:32:18	USE library_system_db	0 row(s) affected
2	21:32:22	USE library_system_db	0 row(s) affected
3	21:33:16	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT CHECK(published_year < 2027)	0 row(s) affected

Limit to 1000 rows

```
1 -- Step 1: Create Database
2 CREATE DATABASE library_system_db;
3 USE library_system_db;
4 -- Step 2: Create Books Table
5 CREATE TABLE Books(
6     book_id INT PRIMARY KEY,
7     title VARCHAR(100) NOT NULL,
8     isbn VARCHAR(20) UNIQUE,
9     published_year INT CHECK(published_year < 2027)
10 );
11 -- Step 3: Create Members Table
12 CREATE TABLE Members(
13     member_id INT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(100) UNIQUE
16 );
```

Output

#	Time	Action	Message
1	21:32:18	USE library_system_db	0 row(s) affected
2	21:32:22	USE library_system_db	0 row(s) affected
3	21:33:16	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT CHECK(published_year < 2027)	0 row(s) affected
4	21:35:36	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email VARCHAR(100) UNIQUE	0 row(s) affected

```

11 -- Step 3: Create Members Table
12 • CREATE TABLE Members(
13     member_id INT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(100) UNIQUE
16 );
17 -- Step 4: Describe Books Table
18 • DESCRIBE Books;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI		
title	varchar(100)	NO			
isbn	varchar(20)	YES	UNI		
published_year	int	YES			

Result 1 x

Output

Action Output

#	Time	Action	Message
✓ 1	21:32:18	USE library_system_db	0 row(s) affected
✓ 2	21:32:22	USE library_system_db	0 row(s) affected
✓ 3	21:33:16	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(...	0 row(s) affected
✓ 4	21:35:36	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email...	0 row(s) affected
✓ 5	21:36:44	DESCRIBE Books	4 row(s) returned

```

17 -- Step 4: Describe Books Table
18 • DESCRIBE Books;
19 -- Step 5: Describe Members Table
20 • DESCRIBE Members;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI		
full_name	varchar(100)	NO			
email	varchar(100)	YES	UNI		

Result 2 x

Output

Action Output

#	Time	Action	Message
✓ 1	21:32:18	USE library_system_db	0 row(s) affected
✓ 2	21:32:22	USE library_system_db	0 row(s) affected
✓ 3	21:33:16	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(...	0 row(s) affected
✓ 4	21:35:36	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email...	0 row(s) affected
✓ 5	21:36:44	DESCRIBE Books	4 row(s) returned
✓ 6	21:37:57	DESCRIBE Members	3 row(s) returned

```

18 DESCRIBE BOOKS;
19 -- Step 5: Describe Members Table
20 DESCRIBE Members;
21 -- Step 6: Insert Records into Books Table
22 INSERT INTO Books VALUES
23 (1,'The Alchemist','9780061122415',1988),
24 (2,'Clean Code','9780132350884',2008),
25 (3,'Atomic Habits','9780735211292',2018);

```

Output

Action Output

#	Time	Action	Message
✓ 2	21:32:22	USE library_system_db	0 row(s) affected
✓ 3	21:33:16	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	0 row(s) affected
✓ 4	21:35:36	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, e...	0 row(s) affected
✓ 5	21:36:44	DESCRIBE Books	4 row(s) returned
✓ 6	21:37:57	DESCRIBE Members	3 row(s) returned
✓ 7	21:38:53	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

```

21 -- Step 6: Insert Records into Books Table
22 INSERT INTO Books VALUES
23 (1,'The Alchemist','9780061122415',1988),
24 (2,'Clean Code','9780132350884',2008),
25 (3,'Atomic Habits','9780735211292',2018);
26 -- Step 7: Display Books Table
27 SELECT * FROM Books;

```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Contents:

	book_id	title	isbn	published_year
▶ 1		The Alchemist	9780061122415	1988
2		Clean Code	9780132350884	2008
3		Atomic Habits	9780735211292	2018
✱	NULL	NULL	NULL	2018

Books 3 x

Output

Action Output

#	Time	Action	Message
✓ 3	21:33:16	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	0 row(s) affected
✓ 4	21:35:36	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, e...	0 row(s) affected
✓ 5	21:36:44	DESCRIBE Books	4 row(s) returned
✓ 6	21:37:57	DESCRIBE Members	3 row(s) returned
✓ 7	21:38:53	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 8	21:39:42	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned

```

26 -- Step 7: Display Books Table
27 • SELECT * FROM Books;
28 -- Step 8: Insert Records into Members Table
29 • INSERT INTO Members VALUES
30 (101,'Harshika','nikshitha@gmail.com'),
31 (102,'Bhanu Priya','bhanupriya@gmail.com'),
32 (103,'Nikshitha','harshika@gmail.com');

```

Output

Action Output

#	Time	Action	Message
✓ 4	21:35:36	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, e...	0 row(s) affected
✓ 5	21:36:44	DESCRIBE Books	4 row(s) returned
✓ 6	21:37:57	DESCRIBE Members	3 row(s) returned
✓ 7	21:38:53	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 8	21:39:42	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓ 9	21:41:51	INSERT INTO Members VALUES (101,'Harshika','nikshitha@gmail.com'), (102,'Bhanu Priya','bhanupriya@gma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

```

33 -- Step 9: Display Members Table
34 • SELECT * FROM Members;

```

Result Grid Filter Rows: Edits Export/Import: Wrap Cell Content:

member_id	full_name	email
101	Harshika	nikshitha@gmail.com
102	Bhanu Priya	bhanupriya@gmail.com
103	Nikshitha	harshika@gmail.com
NULL	NULL	NULL

Members 4 x

Output

Action Output

#	Time	Action	Message
✓ 5	21:36:44	DESCRIBE Books	4 row(s) returned
✓ 6	21:37:57	DESCRIBE Members	3 row(s) returned
✓ 7	21:38:53	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 8	21:39:42	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓ 9	21:41:51	INSERT INTO Members VALUES (101,'Harshika','nikshitha@gmail.com'), (102,'Bhanu Priya','bhanupriya@gma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 10	21:42:46	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned

```

28 -- Step 8: Insert Records into Members Table
29 • INSERT INTO Members VALUES
30 (101,'Harshika','nikshitha@gmail.com'),
31 (102,'Bhanu Priya','bhanupriya@gmail.com'),
32 (103,'Nikshitha','harshika@gmail.com');
33 -- Step 9: Display Members Table
34 • SELECT * FROM Members;
35 -- Step 10: Test Duplicate ISBN
36 • INSERT INTO Books VALUES
37 (4,'Fake Book','9780061122415',2000);

```

Output

Action Output

#	Time	Action	Message
✓ 6	21:37:57	DESCRIBE Members	3 row(s) returned
✓ 7	21:38:53	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 8	21:39:42	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓ 9	21:41:51	INSERT INTO Members VALUES (101,'Harshika','nikshitha@gmail.com'), (102,'Bhanu Priya','bhanupriya@gma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 10	21:42:46	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned

```

30 (101,'Harshika','nikshitha@gmail.com'),
31 (102,'Bhanu Priya','bhanupriya@gmail.com'),
32 (103,'Nikshitha','harshika@gmail.com');
33 -- Step 9: Display Members Table
34 • SELECT * FROM Members;
35 -- Step 10: Test Duplicate ISBN
36 • INSERT INTO Books VALUES
37 (4,'Fake Book','9780061122415',2000);
38 -- Step 11: Test NULL Title
39 • INSERT INTO Books VALUES
40 (5,NULL,'999999999999',2010);

```

Output

Action Output

#	Time	Action	Message
7	21:38:53	INSERT INTO Books VALUES (1,'The Alchemist','9780061122415',1988), (2,'Clean Code','9780132350884',2008), (3,'Atomic Habits','9780735211292',2018);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	21:39:42	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
9	21:41:51	INSERT INTO Members VALUES (101,'Harshika','nikshitha@gmail.com'), (102,'Bhanu Priya','bhanupriya@gmail.com'), (103,'Nikshitha','harshika@gmail.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
10	21:42:46	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
11	21:43:34	INSERT INTO Books VALUES (4,'Fake Book','9780061122415',2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
12	21:49:07	INSERT INTO Books VALUES (5,NULL,'999999999999',2010)	Error Code: 1048. Column 'title' cannot be null

```

22 • INSERT INTO Books VALUES
23 (1,'The Alchemist','9780061122415',1988),
24 (2,'Clean Code','9780132350884',2008),
25 (3,'Atomic Habits','9780735211292',2018);
26 -- Step 7: Display Books Table
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32 (103,'Nikshitha','harshika@gmail.com');
33 -- Step 9: Display Members Table
34 • SELECT * FROM Members;
35 -- Step 10: Test Duplicate ISBN
36 • INSERT INTO Books VALUES
37 (4,'Fake Book','9780061122415',2000);
38 -- Step 11: Test NULL Title
39 • INSERT INTO Books VALUES
40 (5,NULL,'999999999999',2010);
41 -- Step 12: Test Future Publication Year
42 • INSERT INTO Books VALUES
43 (6,'Future Book','888888888888',2030);
44 -- Step 13: Test Duplicate Email
45 • INSERT INTO Members VALUES
46 (104,'Harshika Clone','nikshitha@gmail.com');

```

Output

Action Output

#	Time	Action	Message
10	21:42:46	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned